

Apex POB — Architecture

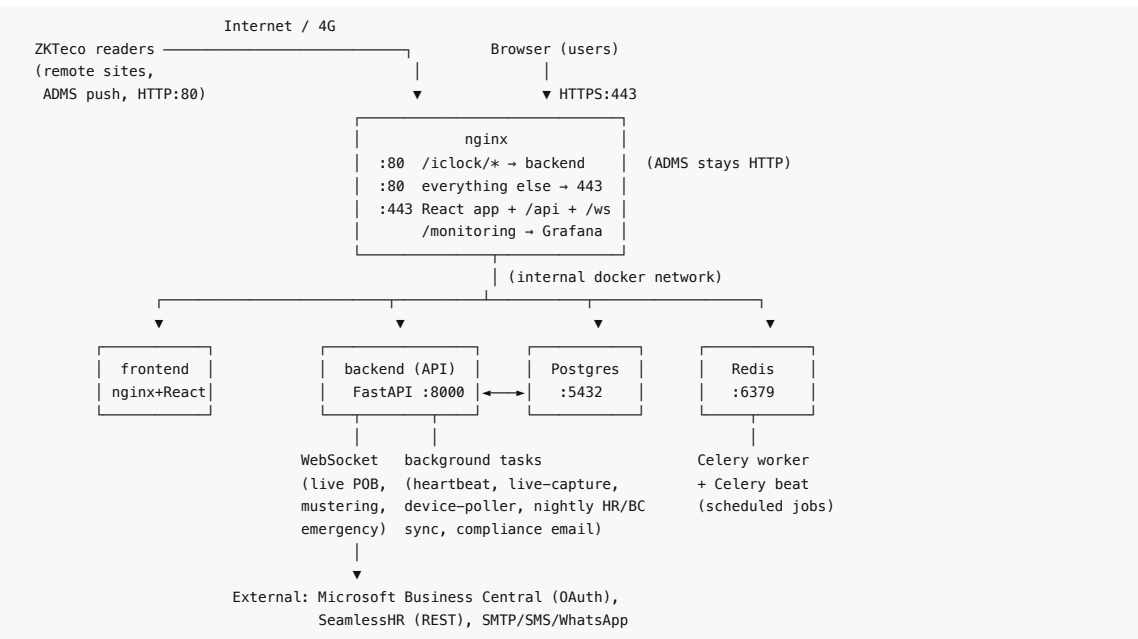
Developer-facing architecture & data-flow reference for the Apex POB platform (Personnel On Board / access control / mustering for oil & gas). Product: **Apex POB** · Operator deployment: **Marconi.ng EPC Limited**

1. Technology stack

Layer	Technology
Frontend	React (CRA + CRACO), Ant Design, Recharts, Leaflet/react-leaflet, React Query
Backend	FastAPI (Python), SQLAlchemy, Pydantic
Database	PostgreSQL 15
Cache / sessions / Celery broker	Redis 7
Background jobs	Celery (worker + beat)
Reverse proxy / TLS	nginx
Containerisation	Docker / docker-compose
Devices	ZKTeco biometric readers via the ADMS push protocol
Monitoring	Prometheus + Grafana + Alertmanager

Note: the original build brief (*Build_guide/Development_Prompt.md*) mentioned Vue.js; the delivered frontend is **React**.

2. Component diagram



3. Reader (ADMS) data flow

- Remote readers are configured with the server's public address (e.g. `http://41.67.137.161` , port **80**, HTTP — readers don't do TLS) in their **Cloud/ADMS** settings. See `project_adms_remote_readers / Build_guide/ZKTECO_INTEGRATION_GUIDE.md` .
- Reader pushes to `/iclock/cdata` (data + heartbeat), polls `/iclock/getrequest` (pending commands), acknowledges via `/iclock/devicecmd` . These are served at the **root path** (`backend/app/api/adms_protocol.py`), no `/api` prefix.
- nginx keeps `/iclock/*` on **HTTP port 80** (everything else redirects to 443).
- `handle_attlog()` routes each punch by the reader's `reader_purpose` :
 - ATTENDANCE** → `iclock_transaction` (T&A) + updates zone occupancy + `personnel.is_onboard` .
 - ACCESS_ENTRY/EXIT** → zone tracking + `access_logs` + `acc_event` , and also a T&A row.
 - MUSTERING** (door in mustering mode) → mustering punch.
- `connection_mode` on the `devices` row controls whether the POB also actively TCP-polls the reader on port 4370 (`direct / both`) or relies purely on ADMS push (`adms`). **Remote readers must be adms** .

4. Occupancy & presence model

- Zone occupancy** and the **POB dashboard** derive "who is where / on board" from the **latest punch** per person (`punch_state IN (0,4)` = checked in; `1 / 5` = out; `255` = auto-detect/presence).

- `personnel.is_onboard` is maintained from punches (check-in → on board; check-out → off board) and is the field the POB dashboard, reports and analytics count.
- Three reads must stay consistent: dashboard (`iclock_transaction` latest punch), per-zone tracking (`zone_personnel_tracking`), and stored `zones.current_personnel_count` (maintained by the punch path). See `project_zone_module` .

5. Key backend modules (`backend/app/api/`)

`adms_protocol` (device protocol), `zones` , `pob_status` , `device_management` / `device_discovery` , `attendance` , `mustering` , `emergency` , `access_control` , `visitor` , `personnel` , `departments` , `payroll` , `reports` , `auth` (+ `mfa` , `sessions`), `bc_integration` , `hr_integration` , `notifications` , `settings` , `audit` .

Services of note: `services/zkteco/{device_heartbeat,device_poller,live_capture}` , `services/{business_central_service,seamlesshr_service,attendance_export,attendance_calculation_service,mustering_service,emergency_s` `core/{security,crypto,rate_limiter,websocket,rbac}` .

6. Real-time (WebSocket)

- Live zone occupancy: `/api/v1/zones/ws` ; access-control events; mustering: `/ws/mustering/...` ; emergency: `/api/emergency/ws/...` ; notifications: `/ws/notifications` .
- All WS URLs are derived client-side from `window.location` (`wss:` on HTTPS) and proxied by nginx (`map $http_upgrade $connection_upgrade`). Never hardcoded `:8000` .

7. Auth & security

- JWT (access/refresh) with **type allowlist** (only `type=access` authenticates) + optional MFA; users live in `auth_user` .
- RBAC middleware; admin/global-admin gating on sensitive endpoints.
- Secrets encrypted at rest (`core/crypto.py`); config validators **hard-fail** on default `SECRET_KEY` / `GLOBAL_ADMIN_PASSWORD` / `LICENSE_SECRET` in production.
- Integration base URLs validated (https-only, SSRF guard). Rate limiting at nginx + app layer. HSTS/CSP in prod nginx.

8. Deployment topology

- **Dev:** `docker-compose.yml` (nginx serves built `frontend-react/build` , HTTP only).
- **Prod:** `docker-compose.prod.yml` + `nginx/conf.d/pob.conf` (TLS on 443, `/iclock` on 80), `.env.prod` , certs in `nginx/certs/` . Services: `postgres` , `redis` , `db-init` , `backend` , `celery-worker` , `celery-beat` , `frontend` , `nginx` , `db-backup` , `prometheus` , `grafana` , `alertmanager` .
- See `DOCKER_DEPLOYMENT.md` , `docs/GOLIVE_HANDOVER.md` .

9. API reference

The authoritative, always-current API spec is **OpenAPI/Swagger** at `/<API_V1_STR>/docs` (non-production) and `openapi.json` . Supplementary: `Build_guide/API_DOCUMENTATION.md` .